

Lab 4: Vibe Coding - Galaga Repair Lab

Pair-Programming Chat History & Implementation Report

1. Overview & Deliverables

This lab focused on debugging and implementing features for the Pygame-based Galaga game using an LLM pair-programming workflow. All 4 tasks were iteratively solved, tested, and committed with separate Git commits.

Task 1: Fix entry/dive Bézier path bug

Found bug in `bezier()` where the third weighted term $3*u*t$ was missing an exponent on t (should be $3*u*t*t$). The sum of Bernstein coefficients was 1.375 instead of 1.0, causing overshooting curves.

Task 2: Implement enemy_tint(kind)

Implemented dynamic enemy palette overriding per wave (emerald, gold, crimson, violet) with safe fallback to `ENEMY_COLORS` for unknown kinds.

Task 3: Implement on_wave_start(wave)

Implemented stage notification banner (e.g. STAGE 1 READY) displayed for 2.0s at each wave onset and updated global wave state.

Task 4: Implement shield_charges(wave)

Granted shield charges per wave $(1 + (\text{wave} - 1) // 3)$ along with visual glowing energy shield aura around ship and HUD counter.

2. Step-by-Step Prompts & Responses

Prompt 1 (Task 1 Bug Fix): Compare the cubic Bézier formula in `bezier()` against standard Bernstein polynomials: $B(t) = (1-t)^3 P_0 + 3(1-t)^2 t P_1 + 3(1-t) t^2 P_2 + t^3 P_3$. Fix the missing exponent on t in the third term.

Response:

Fixed `bezier()` in `game.py`:

```
x = u**3 * p0[0] + 3*u*u*t * p1[0] + 3*u*t*t * p2[0] + t**3 * p3[0]
```

```
y = u**3 * p0[1] + 3*u*u*t * p1[1] + 3*u*t*t * p2[1] + t**3 * p3[1]
```

Verification: $t=0.5$ yields midpoint (150, 150) for symmetric control points.

Prompt 2 (Task 2 Enemy Tint): Implement `enemy_tint(kind)` to return distinct (r, g, b) palettes based on the wave number and enemy kind (boss, red, blue). Return None for unknown kinds.

Response:

Implemented `WAVE_PALETTES` dictionary for waves 1 to 4+ and `enemy_tint(kind)` function returning appropriate RGB tuple with graceful fallback.

Prompt 3 (Task 3 Wave Start): Implement `on_wave_start(wave)` to initialize wave parameters and display a 2-second STAGE banner announcement on screen.

Response:

Implemented `on_wave_start(wave)`, updating `CURRENT_WAVE` and `WAVE_BANNER` with text and timer, integrated in `Game.update` and `Game.draw`.

Prompt 4 (Task 4 Shield Charges): Implement `shield_charges(wave)` to grant shield protection scaling with waves (1 charge at wave 1, +1 every 3 waves) and add visual energy shield aura and HUD counter.

Response:

Implemented `shield_charges(wave)` returning $1 + (\text{wave} - 1) // 3$. Added cyan energy shield aura around player ship and HUD display.

3. Git Commit History for Tasks

Task 1: Fix entry and dive path cubic Bézier curve formula (Task 1)
Task 2: Implement enemy_tint with wave-based dynamic palettes (Task 2)
Task 3: Implement on_wave_start with stage banner announcement (Task 3)
Task 4: Implement shield_charges and player energy shield visuals (Task 4)